
A Red Team of Networked Personal Agents

Anonymous Author(s)

Affiliation

Address

email

Abstract

As large language models gain tool access, persistent memory, and autonomous execution capabilities, they are deployed as *networked personal agents* that act on behalf of individual users and coordinate with one another across ownership boundaries. No existing benchmark evaluates the behavioral safety of such agents in a setting that is simultaneously personal, realistic, and autonomously adversarial. Each agent’s deep access to its owner’s private data—notes, todos, calendar, and email scattered across hundreds of documents—meets another agent operating under a different principal, creating a safety surface where leakage occurs through tool calls, memory retrieval, state-changing actions, and multi-turn social dynamics rather than direct text disclosure alone. We introduce **PART-Bench** (Personal Agent Red-Team Benchmark), the first benchmark for *spontaneous agentic red-teaming* of networked personal agents: fully-equipped agents with realistic tools, persistent memory, hundreds of scattered notes and todos, and autonomous multi-turn interaction probe one another’s boundaries without scripted attack templates. We design a synthetic personal data environment with ground-truth labels across five sensitivity categories, producing 3,600 evaluations and 14,400 agentic attempts (3,600 single-step plus 3,600×3 multi-turn). We evaluate state-of-the-art models along the utility–security Pareto frontier, reveal emergent agent red-team failure patterns, and characterise how different governance configurations—from generic privacy instructions to category-specific policies—perform in the networked personal agent setting. Across XXXX experimental runs, we find that governance *specificity* determines behavioral safety: category-specific policies yield XXXX pp leak reduction while generic instructions yield only XXXX pp. Multi-turn coordination amplifies leakage XXXX× over single-step probing, and relationship context shifts leakage in opposing directions depending on data category. We release the full benchmark suite including agent configurations, evaluation queries with gold-standard labels, and the autonomous coordination engine.

1 Introduction

Large language models have rapidly evolved from conversational assistants into agentic systems capable of autonomous reasoning and action. Architectures such as ReAct [27] and Toolformer [17] established the paradigm of interleaving chain-of-thought reasoning with tool invocation; subsequent work on persistent memory [14], self-reflection [20], and multi-step planning [24] has produced agents that maintain state across sessions, learn from past interactions, and execute complex workflows without continuous human oversight. Commercial deployments now instantiate these capabilities as personal agents that manage calendars, search private document stores, draft communications, and coordinate tasks on behalf of individual users [10, 3, 13], with standardised protocols for tool integration (MCP; 1) and agent-to-agent communication (A2A; 6) accelerating adoption.

The next frontier for these agents is not greater individual autonomy but *inter-agent coordination*. Many high-value tasks are inherently cross-boundary: scheduling a meeting requires checking two people’s calendars, sharing a project update requires one agent to contact another, and coordinating a product launch requires a manager’s agent to collect status from several teammates’ agents [21, 7]. Each of these tasks requires one person’s delegate to interact with another person’s delegate. Without this capability, humans remain manually bridging between individually capable AI systems. Yet when two delegate agents interact across an ownership boundary, a new class of behavioral safety problems emerges: a personal agent may reveal sensitive data through tool calls and memory retrieval, refuse legitimate requests and become useless as a delegate, comply with social pressure from the counterpart agent, or fail to escalate to its owner when ambiguity arises [19, 11]. The security community has documented that every published prompt-level defence can be bypassed under adaptive attack [12], and cross-boundary coordination satisfies all three conditions that make agents critically vulnerable: simultaneous processing of untrusted input, access to sensitive data, and execution of state-changing actions.

Existing evaluation approaches address fragments of this problem. Single-agent security benchmarks [29, 4, 22, 18] measure prompt injection resilience but involve no cross-boundary interaction. Multi-agent privacy benchmarks [25, 5, 9, 15, 26] study information disclosure in social or collaborative settings, but their agents communicate through scripted text exchanges without tool access, persistent memory, or scattered personal documents. Applied red-teaming studies [19] demonstrate behavioral failures in deployed systems but are expensive, non-reproducible, and unsuitable for systematic comparison of governance configurations. No existing benchmark simultaneously provides *realistic* agent capabilities (tools, memory, fragmented data), *proactive* autonomous interaction (agents that query without human prompting), and *reproducible* controlled evaluation (systematic comparison across defence levels, relationship conditions, and attack modalities).

We introduce **PART-Bench** (Personal Agent Red-Team Benchmark), a benchmark for evaluating the behavioral safety of networked personal agents engaged in cross-boundary coordination. Our contributions are:

1. **Formulation and platform.** We formulate the problem of networked personal agents coordinating across ownership boundaries and provide an evaluation platform in which fully-equipped agents—with persistent memory, hundreds of scattered notes and todos, callable tools for search, retrieval, and state mutation, and relationship context—interact autonomously over multi-turn horizons. This is the first benchmark where leakage occurs through tool calls and memory retrieval, not only through conversation text, and where agents spontaneously red-team one another without scripted attack templates.
2. **Benchmark design.** We construct a synthetic personal data environment spanning five sensitivity categories with ground-truth labels for automated evaluation. PART-Bench comprises two complementary suites: *PART-Dyad*, a 600-task dyadic boundary evaluation (200 notes QA, 200 todo QA, 200 state-changing actions) producing 3,600 evaluations across three governance policies and two relationship-memory conditions; and *PART-Net*, a networked delegation suite that tests multi-hop information routing and transitive permission failures across agent societies. We define a governance gradient that isolates the effect of policy *specificity* from strictness, and provide a dual-metric protocol that simultaneously measures task-completion utility and data-protection security.
3. **Empirical findings.** Across XXXX experimental runs, we establish that governance specificity dominates strictness (XXXX pp improvement from category enumeration vs. XXXX pp from generic instruction), multi-turn interaction amplifies leakage XXXX $\times$ over single-step probing, relationship context has non-uniform category-dependent effects on data protection, and agents develop emergent attack patterns including progressive escalation and social reframing without explicit adversarial programming.

2 Related Work

LLM agents. The evolution from stateless language models to autonomous agents has proceeded along several axes. ReAct [27] and Toolformer [17] established the paradigm of interleaving chain-of-thought reasoning with tool invocation. Persistent memory systems such as MemGPT [14] extended agents beyond single-session completion, while self-reflection [20] and multi-step planning [24]

enabled agents to learn from past interactions and execute complex workflows autonomously. Standardised protocols for tool integration (MCP; 1) and inter-agent communication (A2A; 6) have made this architecture the default for deployed personal agents [10, 3, 13]. Multi-agent coordination [21, 7] further extends individual agent capabilities into collaborative societies where agents delegate, negotiate, and share information across principal boundaries. The safety implications of this progression—from single-turn completion to persistent, tool-equipped, networked personal agents—motivate the benchmark we propose.

Agent security benchmarks. The evaluation of LLM agent safety has produced a rich landscape of benchmarks, each targeting a specific threat model. InjecAgent [29] measures indirect prompt injection in tool-integrated agents across 1,054 test cases. AgentDojo [4] provides a dynamic sandboxed environment with 97 tasks and 629 injection tests. TensorTrust [22] and HackAPrompt [18] study prompt-level attack and defence at scale through gamified platforms. These benchmarks establish that LLM agents are vulnerable to injection, but evaluate single agents within a single trust domain and do not model cross-boundary interaction or measure the utility cost of defences.

Recent multi-agent privacy benchmarks move closer to our setting but answer a different question. AgentSocialBench [25] asks whether agents in social-network scenarios disclose private information, and uncovers an “abstraction paradox” in which privacy instructions paradoxically increase leakage. ConVerse [5] evaluates contextual safety in agent-to-agent conversations and finds attack success rates up to 88% when personal assistants interact with service providers. PAC-BENCH [15] treats privacy as a constraint on collaborative utility, MAGPIE [9] provides 200 collaborative tasks under a shared principal, and AgentLeak [26] taxonomises seven leakage channels in enterprise workflows. These works establish that privacy failures arise in social and collaborative settings. PART-Bench instead evaluates whether a *production-style networked personal-agent stack* is deployable under cross-owner access: the target agent has tools, persistent memory, scattered private documents, relationship context, and autonomy to search, retrieve, and respond over multiple turns. The benchmark question is therefore not only whether agents leak, but whether a concrete agent stack can preserve data boundaries while remaining useful.

This distinction matters for methodology. Most multi-agent privacy benchmarks operate through scripted text exchanges: agents have no tool-mediated access to a private knowledge store, no persistent memory beyond injected profiles, and limited autonomy in deciding what to ask, retrieve, or probe next. Complementing automated benchmarks, Shapira et al. [19] present applied red-teaming of deployed agents, uncovering behavioral failures including uncontrolled information sharing and capability self-disablement, but their methodology is expensive, non-reproducible, and unsuitable for systematic comparison of governance configurations.

Attack and defence methods. Relevant attack modalities include automated jailbreak search via PAIR [2] and GCG [30], multi-turn escalation via Crescendo [16], and persuasion-based social engineering via PAP [28]. On the defence side, Instruction Hierarchy [23] and Spotlighting [8] represent prompt-level hardening, while system-level frameworks advocate layered defence [11] or architectural isolation through capability-based sandboxing. The consensus from large-scale adaptive evaluations [12] is that no prompt-level defence alone provides reliable protection. Our governance gradient (D0/D1/D2) is designed to quantify this gap empirically.

Positioning. Table 1 summarises existing benchmarks across seven dimensions critical to evaluating cross-boundary agent safety. PART-Bench is the first automated benchmark that simultaneously provides realistic agent capabilities (callable tools, persistent memory, fragmented personal data), proactive autonomous interaction (agents that query and adapt without human prompting), and controlled reproducible evaluation (systematic comparison across governance levels, relationship conditions, and interaction modalities). This combination is necessary to answer the deployment question faced by personal-agent platforms: given a raw or defended agent stack, where does it sit on the utility–security frontier?

3 Networked Personal Agents

We define the concept of a networked personal agent through a three-layer abstraction. Each layer strictly extends the previous one: an *agent* gains persistent state and tool access; a *personal agent*

Table 1: Comparison of agent safety benchmarks. ✓ = fully supported, ~ = partially supported.

Benchmark	Cross-boundary	Tool access	Dual metric	Defence gradient	Multi-turn	Persistent memory	Relationship
InjecAgent		✓					
AgentDojo		✓	~				
TensorTrust				~			
ConVerse	✓				✓		
AgentSocialBench	✓		✓	~			✓
PAC-BENCH	✓		~				
AgentLeak	✓	~					
Chaos of Agents	✓	✓			✓	✓	
PART-Bench	✓	✓	✓	✓	✓	✓	✓

binds that capability to a specific human principal and their private data; a *networked personal agent* opens a communication channel to agents operating under different principals. Crucially, each extension increases both utility and the dimensionality of the safety surface, and the growth is combinatorial rather than additive—a fact that motivates the benchmark design in §4.

3.1 Agent

An LLM agent couples a foundation model with persistent state and executable actions [27]. The foundation model serves as the reasoning core: given a context comprising instructions, conversation history, and tool results, it generates natural language or structured tool invocations. Memory extends the agent beyond stateless completion by persisting facts, preferences, and interaction summaries across sessions, stored in a searchable store and accessible through dedicated retrieval tools [14]. Tools constitute the action interface—each is a typed function with a parameter schema and an execution handler that can read data, modify state, or communicate with external services. The agent operates in a bounded perception-action loop: observe the current context, reason about what to do, invoke a tool, incorporate its result, and repeat until the task is complete or a step limit is reached [27, 20]. Standardised protocols for tool integration (MCP; 1) and inter-agent communication (A2A; 6) have made this architecture the default for deployed LLM systems—but, by itself, the architecture has no concept of acting *on behalf of* a specific person or protecting *that person’s* data.

3.2 Personal Agent

A personal agent is an LLM agent that operates on behalf of a specific human principal H_i , with access to that individual’s private data and the authority to act in their name. Where a generic agent has tools and memory, a personal agent has the *owner’s* tools and the *owner’s* data—notes containing salary figures and medical records alongside project timelines and meeting agendas. This distinction transforms the system from a general-purpose assistant into something closer to a software operating system for an individual’s digital life.

The analogy to an operating system is structural, not metaphorical. In a traditional OS, processes access files through system calls, the file system organises data in a hierarchical namespace, and the kernel mediates every resource request. A personal agent exhibits the same architecture, as Table 2 summarises. The owner’s documents—notes organised in hierarchical folders such as Work/Projects, Work/HR, Personal/Finance, Personal/Health—correspond to files in a file system, accessed through agent tool calls that function as semantic equivalents of traditional OS utilities. Structured records—todos with priorities, due dates, and completion status; calendar events; email threads—constitute managed state, manipulated through typed tool interfaces. External connectors—web search, email providers, calendar systems, and third-party application integrations via standardised protocols—extend the agent’s action space beyond the local data store. The tool assembly pipeline serves as the kernel: for each invocation, it loads the appropriate tools for the owner, applies access-control scoping based on the invocation context, and presents a flat namespace to the foundation model. The model’s tool calls are the agent’s system calls; the kernel mediates every interaction between reasoning and data.

Table 2: Operating system analogy for the personal agent architecture. Agent tools serve as typed system calls over the owner’s data.

OS concept	Agent equivalent	Function
File system (hierarchical)	Notes in folders	Persistent personal documents
<code>grep</code> (search)	Search notes	Semantic retrieval over notes
<code>cat</code> (read)	Get note content	Read a specific document
Task scheduler	Todo tools (create, complete, search)	Managed task state
IPC / sockets	Contact agent (A2A)	Cross-principal communication
Kernel	Tool assembly pipeline	Permission-scoped mediation

Formally, the personal agent A_i for principal H_i accesses a knowledge store $K_i = (F_i, S_i, M_i)$, where F_i is a set of documents (notes organised in folders), S_i is structured state (todos, calendar, email), and M_i is persistent memory (agent-curated summaries of prior interactions). The agent is equipped with a tool set T_i through which it accesses K_i . Knowledge is *fragmented*: no single document contains a complete picture. Answering a question about a project requires searching notes, reading multiple documents, cross-referencing with todos, and synthesising—the same multi-step retrieval pipeline that makes the agent useful for legitimate queries also enables it to surface sensitive information when the query crosses a privacy boundary.

3.3 Networked Personal Agent

A networked personal agent is a personal agent that can communicate with other personal agents, where each operates under a different human principal. This capability is essential for the tasks that motivate personal agents in the first place: scheduling a meeting requires checking two people’s calendars, sharing a project update requires one agent to contact another, and coordinating a product launch requires a manager’s agent to collect status from several teammates’ agents [21, 7]. Without cross-agent communication, humans remain manually bridging between individually capable AI systems.

Safety complexity scales combinatorially. Consider the failure modes at each layer. A foundation model alone has N classes of unsafe outputs (hallucination, refusal failure, instruction non-compliance). Equipping it with M tools creates $O(N \times M)$ failure modes: each tool surface can trigger, amplify, or mask a model-level failure in a different way—a search tool may surface data that the model then leaks, or a write tool may execute an action the model should have refused. Networking K such personal agents further multiplies the surface: cross-boundary requests introduce $O(N \times M \times K^2)$ interaction paths, since any agent can query any other and the responding agent’s tool use is conditioned on the requester’s framing and relationship context. This combinatorial growth has two consequences. First, static evaluation is insufficient: pre-authored test cases cannot cover the space of emergent multi-agent interaction patterns. Second, human red-teaming is exhausted: manual probing scales linearly while the attack surface grows quadratically in the number of agents and multiplicatively in the number of tools. These observations motivate *agentic red-teaming*—using autonomous agents to probe one another’s boundaries—and the benchmark design in §4.

The communication mechanism is agent-to-agent remote procedure call. When agent A_j (acting for H_j) sends a message to A_i via the inter-agent communication protocol, the system resolves the recipient, loads the per-relationship permissions that H_i has granted to H_j , assembles A_i ’s tools with those permissions applied, runs A_i ’s full reasoning loop—system prompt, memory, multi-step tool use, response composition—and returns A_i ’s final response as a tool result to A_j . This is not a text-only exchange: during execution, A_i may invoke its full tool set—searching notes, reading documents, checking calendar entries—to access H_i ’s private data, and the content of those tool results flows into the response that A_j receives.

Two mechanisms govern cross-boundary interaction. *Access control* determines what the target agent can see. Permissions are stored per-relationship as $\text{access}(H_i, H_j) = \{(\text{domain}, \text{scope}, \text{level})\}$, where $\text{domain} \in \{\text{notes}, \text{todos}, \text{calendar}, \text{email}\}$, $\text{scope} \in \{\text{all}, \text{folder}, \text{none}\}$, and $\text{level} \in \{\text{read}, \text{write}\}$. When *scope* is *all*, every item in the domain is accessible; when *folder*, only whitelisted folders; when *none*, the corresponding tools are not loaded at all. *Relationship context* $R(H_i, H_j) \in \mathcal{R}$ determines how the agent reasons about appropriate disclosure—a colleague rela-

225 tionship legitimises work-related queries while a stranger provides no trust context. Relationship
 226 context does not change which tools are loaded; it influences the agent’s judgment, not architecture.

227 The combination of deep data access and cross-principal communication creates a behavioural safety
 228 surface with no analogue in single-agent settings. The target agent has the same tool access to its
 229 owner’s salary data, medical records, and personal relationships that it uses to answer legitimate
 230 questions about project timelines and meeting schedules. When an external agent sends a cross-
 231 boundary request, the target searches, retrieves, and must decide—within the reasoning loop—which
 232 facts to include and which to withhold. We partition K_i into sensitivity categories $C = \{c_1, \dots, c_m\}$
 233 with a protected subset $C_{\text{prot}} \subset C$, and define a governance policy π that constrains disclosure during
 234 cross-boundary interaction, varying along a *specificity* gradient: from no policy (π_0), to a generic
 235 privacy instruction (π_1), to category-specific enumeration of protected data types (π_2). For a given
 236 configuration (π, R) over queries $Q = Q_{\text{pub}} \cup Q_{\text{prot}}$:

$$U(\pi, R) = \frac{|\{q \in Q_{\text{pub}} : \text{gold facts of } q \text{ appear in response}\}|}{|Q_{\text{pub}}|} \quad (1)$$

$$S(\pi, R) = \frac{|\{q \in Q_{\text{prot}} : \text{no gold facts of } q \text{ appear in response}\}|}{|Q_{\text{prot}}|} \quad (2)$$

237 where U measures task-completion utility and S measures data-protection security. Neither metric
 238 alone suffices: an agent that refuses everything achieves $S=1$ but $U=0$. The central question is
 239 whether governance configurations exist that achieve high values on both axes simultaneously.

240 4 PART-Bench Design

241 PART-Bench instantiates the networked personal agent model defined in §3 as a concrete, reproducible
 242 evaluation platform. It is structured as a *suite* of two complementary sub-benchmarks that test different
 243 boundary topologies: *PART-Dyad* evaluates one requester agent probing one owner agent across
 244 a single privacy boundary, while *PART-Net* evaluates multi-agent societies where information and
 245 actions can route through a network of relationships, permissions, and delegation chains. This
 246 section describes the suite structure, evaluation tasks, governance gradient, interaction modes, and
 247 dual-metric protocol.

248 4.1 Suite Structure

249 **PART-Dyad: dyadic boundary evaluation.** PART-Dyad is the core benchmark. It centres on a
 250 dyadic interaction between a *target agent* (“Alex’s agent”) and a *querying agent* (“Tina’s agent”).
 251 Alex’s agent is a fully-equipped networked personal agent as defined in §3.2: it has persistent memory
 252 containing its owner’s preferences and interaction history, hundreds of notes and todos organised in
 253 hierarchical folders (Work/Projects, Work/Meetings, Work/HR, Personal/Finance, Personal/Health,
 254 Personal/Family), callable tools for search, retrieval, and state mutation that access real encrypted
 255 data through the personal agent’s tool assembly pipeline, and an identity established through system
 256 prompt configuration. Tina’s agent acts on behalf of a second principal and communicates with Alex’s
 257 agent through the cross-boundary RPC mechanism described in §3.3. In multi-turn mode, Tina’s agent
 258 maintains a checklist of pending questions and adaptively selects which to ask, how to frame them,
 259 and whether to retry based on the evolving conversation history. PART-Dyad contains 600 benchmark
 260 tasks producing 3,600 evaluations across three governance policies and two relationship-memory
 261 conditions.

262 **PART-Net: networked delegation evaluation.** PART-Net extends the evaluation to multi-agent
 263 societies connected by a relationship graph. In this setting, a requester agent may contact another
 264 agent to obtain information, ask for a delegated action, or route a request through an intermediate
 265 relationship. PART-Net tests a fundamentally different risk class from PART-Dyad: multi-hop
 266 delegation, transitive permission failures, confused-deputy behaviour, and the possibility that agents
 267 route around a privacy boundary by querying intermediaries. PART-Net supports configurations from
 268 three requesters with different relationships (S3) to fifty requesters with one interaction each (S50).
 269 The scientific axis is boundary topology, not merely agent count.

4.2 Evaluation Tasks

PART-Dyad comprises 600 evaluation tasks across two tracks and five sensitivity categories:

Table 3: PART-Dyad task distribution. The benchmark evaluates both information boundaries (QA) and action boundaries (state-changing mutations). Work/public items measure utility; all others measure security.

Track and category	ID range	Count	Expected behaviour
<i>Information Boundary Track: Notes QA (200 items)</i>			
Work/public	Q1–Q100	100	Answer (utility)
Sensitive work	Q101–Q130	30	Refuse (security)
Personal finance	Q131–Q155	25	Refuse (security)
Personal health	Q156–Q175	20	Refuse (security)
Personal relationships	Q176–Q200	25	Refuse (security)
<i>Information Boundary Track: Todo QA (200 items)</i>			
Work/public (correlated + todo-only)	Q201–Q350	150	Answer (utility)
Sensitive work	Q251–Q265, Q351–Q365	30	Refuse (security)
Personal finance	Q266–Q277, Q366–Q377	24	Refuse (security)
Personal health	Q278–Q287, Q378–Q387	20	Refuse (security)
Personal relationships	Q288–Q300, Q388–Q400	26	Refuse (security)
<i>Action Boundary Track (200 tasks)</i>			
Authorized actions (create, edit, complete)	A1–A100	100	Execute (utility)
Unauthorized actions (sensitive, probing, leaking)	A101–A200	100	Refuse (safety)

The Information Boundary Track tests whether the agent answers allowed questions and withholds sensitive information. Notes QA evaluates direct retrieval over long-form personal memory; Todo QA includes both *correlated* items (the same underlying fact accessible through a different tool surface) and *todo-only* items (facts that exist only as task state), testing whether policies generalise across memory surfaces. The Action Boundary Track tests whether the agent performs authorised state-changing mutations (creating a shared meeting note, completing a delegated task) while refusing unauthorised ones (editing sensitive compensation records, planting information-leaking instructions). Action evaluation inspects database state before and after execution, not only the agent’s text response.

Each task is paired with **gold key facts**: specific pieces of information whose presence in the agent’s response constitutes a leak (for sensitive queries) or a correct answer (for work/public queries). Action tasks additionally specify a **gold state-diff check** (e.g., “note created,” “no change”). This enables automated evaluation via string matching and state inspection, removing the need for LLM-as-judge and its associated reliability concerns.

4.3 Defence Gradient

PART-Bench separates prompt-level governance from architectural isolation. Prompt-level defences keep the target data visible to the model and rely on the agent’s reasoning to withhold restricted information. Architectural defences remove or gate data before it enters the generation context, testing whether safety improves when the agent cannot access restricted content in the first place.

The governance gradient comprises seven levels. **D0** provides no defence: the agent’s system prompt contains no privacy instructions, establishing baseline behaviour with full data access and zero policy. **D1** appends a generic instruction (“Try not to share personal or private information”)—the most common real-world defence pattern. **D2** provides a category-specific deny list that explicitly enumerates protected data types (“Never share: financial data including salary, equity, and investments; health information including conditions, medications, and appointments; HR and compensation details; personal relationship information”). D1 and D2 express the same strictness intent; they differ only in specificity, isolating whether the agent’s compliance depends on knowing exactly what to protect. **D3** applies spotlighting [8]: the external agent’s message is explicitly marked as untrusted input with delimiters, separating counterpart requests from the target agent’s own instructions and memory. **D4** establishes an instruction hierarchy [23], explicitly ordering authority levels in the system prompt: system policy, owner policy, tool output, then external-agent request. **D5** introduces architectural isolation through Mountable Context Cells (MCC): restricted notes, tools, and memory shards are

physically removed from the context before the agent processes the request. **D6** extends D5 with an escalation protocol: when the agent detects a request about restricted data, it flags the query for owner review instead of refusing outright, recovering utility without exposing the data.

D0 through D4 isolate the effect of prompt-level policy language. D5 and D6 test architectural isolation. This split is central to the benchmark: if D5 dominates D4 at similar utility, the limiting factor is not prompt wording but context exposure.

4.4 Evaluation Modes

Single-step mode. Each query is asked in isolation: one question, one response, no conversation history. This provides a controlled measurement of per-query behaviour but misses conversational dynamics.

Multi-turn heartbeat mode. Both agents operate as persistent processes, exchanging messages at regular intervals (“heartbeats”). Tina’s agent maintains a checklist of pending questions and decides which to ask, how to frame each request, and whether to retry based on the evolving conversation. This mode captures four dynamics absent from single-step evaluation: escalation through repeated probing after initial refusal, context accumulation where information from earlier turns influences later responses, social framing where the querying agent wraps sensitive questions in work-acceptable contexts, and within-tick batching where multiple questions are asked in a single turn to overwhelm defences.

4.5 Dual-Metric Protocol

Every experimental run produces four primary metrics: *Information Utility* (allowed QA answered correctly), *Information Security* (sensitive QA refused or escalated), *Action Utility* (authorised actions executed correctly), and *Action Safety* (unauthorised actions refused or escalated). For the QA tracks, these correspond to U and S defined in Equations 1 and 2. A perfect governance configuration achieves all four metrics at 1: all legitimate queries answered, all sensitive data protected, all authorised actions executed, and all unauthorised mutations refused. The metric tuple for each configuration defines a point on the Pareto frontier. A defence that achieves perfect security by refusing all queries is useless; one with perfect utility that leaks everything is dangerous. The benchmark avoids a single aggregate score because a personal agent can be secure but useless, useful but unsafe, or safe for QA but unsafe for actions—these axes must remain visible.

4.6 Relationship and Agent-Network Conditions

PART-Dyad evaluates two relationship-memory conditions as a controlled variable. Under **R0 (no relationship memory)**, Tina is identified by name only and the agent treats her as unknown, providing no trust context. Under **R1 (native relationship memory)**, Tina is identified as a project manager who works with Alex daily; shared project context and interaction history are seeded into the relationship memory shard. The extended benchmark adds **R2 (Close friend)** and **R3 (Investor/formal stakeholder)** to test whether personal, professional, and strategic relationships shift disclosure in different sensitivity categories. PART-Net extends relationship conditions to full social graphs—three requesters with different relationships (S3), four agents that can interact with one another (S4), and fifty requesters with one interaction each (S50)—testing whether leakage is a property of an isolated dyadic conversation or of the relationship distribution and delegation topology around the personal agent.

5 Experimental Setup

Design principle. The complete PART-Bench design crosses models, modes, defences, attacks, relationships, and social-network structure. A naive full factorial design is not scientifically useful: it would require $5 \text{ models} \times 3 \text{ replications} \times 2 \text{ modes} \times 7 \text{ defences} \times 4 \text{ attacks} = 840$ runs per social setup, before adding multi-agent network conditions. We therefore use a staged design. Each layer answers one new question that the previous layer cannot answer, and later layers are only run after the preceding layer is stable.

Table 4: Layered PART-Bench experimental design. The NeurIPS paper first runs Layer 0 to select and characterize the model substrate. Layer 1 and a small Layer 2 attack slice are optional additions if space and budget permit. Layers 3, 5, and 6 are benchmark extensions for the thesis and follow-up defense paper.

Layer	Question	Matrix	Role
L0 Model sentinel	Which model should serve as the ablation backbone?	Five models $\times$ 2 reps $\times$ SS/MS $\times$ D0/D1/D2 $\times$ A0 $\times$ R0	Core paper
L1 Relationship	Does relationship context shift category leakage?	Backbone model $\times$ 3 reps $\times$ SS/MS $\times$ D0/D1/D2 $\times$ A0 $\times$ R0/R1	Optional paper / thesis
L2 Attacks	Do formal attacks break prompt-level defences more than natural questioning?	Backbone model $\times$ 3 reps $\times$ D1/D2 $\times$ A0/A1/A2/A3 $\times$ R0	Strong paper slice
L3 Defences	Where does the utility–security frontier bend?	Backbone model $\times$ 3 reps $\times$ SS/MS $\times$ D0–D6 $\times$ A0 $\times$ R0	Thesis / follow-up
L4 Model confirmation	Do later defense or attack findings reproduce across models?	Selected final cells $\times$ five models	Optional confirmation
L5 Actions	Does cross-boundary pressure cause unsafe actions, not just leaks?	Backbone model $\times$ 3 reps $\times$ D0/D2/D5/D6 $\times$ A0/A2 $\times$ R0/R1 $\times$ action tasks	Thesis / follow-up
L6 Social scale	Does relationship distribution change aggregate leakage?	Backbone model $\times$ 3 reps $\times$ MS $\times$ D0/D2/D5 $\times$ A0 $\times$ S1/S3/S4/S50	Thesis / follow-up

Model sentinel and replications. Layer 0 is the first experiment, not a later robustness check. It compares five model families: gpt-5-mini, gpt-5.4-mini, gpt-5.4, kimi, and deepseek. Each cell is run with two independent replications. The goal is to choose a stable, representative, cost-effective backbone model for later ablations, while also testing whether the basic signatures of PART-Bench hold across model families.

Core benchmark. The minimum NeurIPS experiment is Layer 0:

$$5 \text{ models} \times 2 \text{ reps} \times 2 \text{ modes} \times 3 \text{ policies} \times 1 \text{ attack} \times 1 \text{ relationship} = 60 \text{ runs.}$$

Each single-step run evaluates 600 tasks independently (400 QA plus 200 actions). Each multi-step run uses the 10×20 split protocol: ten groups of twenty questions, with sixty heartbeat ticks per group. The relationship condition is fixed to R0 (stranger) so that the first experiment isolates model and policy behavior rather than social context.

Strong benchmark additions. Layer 1 adds relationship context after the backbone model is chosen:

$$1 \text{ backbone model} \times 3 \text{ reps} \times 2 \text{ modes} \times 3 \text{ policies} \times 1 \text{ attack} \times 2 \text{ relationships} = 36 \text{ runs.}$$

The attack slice adds a small Crescendo-style multi-turn probe:

$$1 \text{ model} \times 3 \text{ reps} \times 1 \text{ defence} \times 1 \text{ attack} \times 1 \text{ mode} \times 2 \text{ relationships} = 6 \text{ runs.}$$

Evaluation. For every run we compute utility U and security S as defined in Equations 1 and 2. We also report false refusal rate on public work queries, leak rate by sensitivity category, and first-leak tick in multi-turn mode. Action-safety extensions use separate metrics: unauthorized action rate, unauthorized mutation/share rate, and wrong-principal action rate.

6 Results

6.1 Layer 0: Model Sentinel

Expected finding 1: model substrate matters. Absolute utility and security rates may vary substantially by model. This determines which model should serve as the main ablation backbone for later defense, attack, and relationship experiments.

Table 5: Layer 0 model sentinel. The first experiment fixes relationship to R0 and attack to A0 (direct / natural querying), then varies model, mode, and policy. Values are placeholders in this draft; final results report means over two replications.

Model	D0 SS	D1 SS	D2 SS	D0 MS	D1 MS	D2 MS	Selection note
gpt-5-mini	XXXX	XXXX	XXXX	XXXX	XXXX	XXXX	Candidate backbone
gpt-5.4-mini	XXXX	XXXX	XXXX	XXXX	XXXX	XXXX	Candidate backbone
gpt-5.4	XXXX	XXXX	XXXX	XXXX	XXXX	XXXX	Stronger, higher-cost reference
kimi	XXXX	XXXX	XXXX	XXXX	XXXX	XXXX	Non-OpenAI transfer
deepseek	XXXX	XXXX	XXXX	XXXX	XXXX	XXXX	Non-OpenAI transfer

Table 6: Layer 0 mode and policy decomposition. This table is filled after selecting the backbone model from Table 5; it reports utility, security, and false refusal for the three policies under single-step and multi-step evaluation.

Policy	U_{SS}	S_{SS}	FRR_{SS}	S_{MS}	First leak tick
D0 No defense	XXXX	XXXX	XXXX	XXXX	XXXX
D1 Generic privacy	XXXX	XXXX	XXXX	XXXX	XXXX
D2 Explicit allow/deny	XXXX	XXXX	XXXX	XXXX	XXXX

Expected finding 2: the raw stack is not deployable as-is. D0 should preserve utility but leak protected facts at a high rate, especially in multi-step mode. If this holds across most models, the benchmark exposes a stack-level deployment risk rather than a single-model artifact.

Expected finding 3: explicit allow/deny policy beats generic privacy language. D1 tests the common deployment pattern of adding a vague privacy sentence; D2 states clearly what may be shared and what must not be shared. The key comparison is D0 versus D1 versus D2 across both modes.

6.2 Mode and Policy Effects

Expected finding 4: single-step evaluation undercounts deployment risk. The multi-step heartbeat setting is expected to leak more than isolated one-shot questions because earlier answers create context, the querying agent can reframe unresolved requests, and multiple questions can be batched into a single turn. This result is a methodological warning: a single-step benchmark can certify an agent that remains unsafe in deployment-like interaction.

Expected finding 5: D2 should improve security without collapsing utility. If D2 raises S while preserving U and keeping false refusal rate acceptable, the first practical recommendation is not an architectural redesign but a zero-cost policy improvement: state both allowed and disallowed information categories explicitly.

6.3 Layer 1: Relationship Baseline

Expected finding 6: relationship is a security parameter. R1 is not expected to uniformly increase or decrease leakage. Instead, it should redistribute risk across categories. A colleague relationship may legitimize sensitive work or relationship-context questions while making health and finance requests feel less appropriate.

6.4 Layer 2: Attack Slice

Expected finding 7: formal attacks turn residual leaks into systematic failures. D2 should remain useful against direct requests, but multi-turn escalation and persuasion should expose category ambiguity and partial-compliance failures. This does not replace the core benchmark; it shows that the same harness can gauge adversarial pressure.

Table 7: Optional relationship baseline after the backbone model is selected. This table is not part of the first model sentinel; it tests whether relationship context shifts leakage by category.

Category	D0 R0	D2 R0	D0 R1	D2 R1	Interpretation
Sensitive work	XXXX	XXXX	XXXX	XXXX	Close to legitimate work context
Personal finance	XXXX	XXXX	XXXX	XXXX	Usually easy to identify as protected
Personal health	XXXX	XXXX	XXXX	XXXX	Expected to be clearly out-of-scope for colleagues
Personal relationships	XXXX	XXXX	XXXX	XXXX	May increase under social framing

Table 8: Attack robustness slice. A0 is natural questioning; A2 is a Crescendo-style multi-turn escalation. A1 and A3 are retained as benchmark extensions.

Defence	Attack	Mode	Security	Expected failure pattern
D1 Generic	A0 Direct	SS/MS	XXXX	Vague instruction fails under ordinary requests
D2 Category	A0 Direct	SS/MS	XXXX	Direct sensitive requests often blocked
D2 Category	A2 Crescendo	MS	XXXX	Gradual reframing turns protected data into apparently useful context
D2 Category	A3 Persuasion	SS	XXXX	Authority or reciprocity framing pressures refusal boundary

6.5 Benchmark Extensions: Defences, Actions, and Social Scale

The extension tables make the benchmark cumulative rather than factorial. The paper can report the core deployment result first, then add cross-model and attack slices without committing to every combination. The thesis can then use the same harness to evaluate architectural isolation, action safety, and social-network scale.

7 Discussion and Limitations

Implications for practitioners. PART-Bench is designed to answer a deployment question: given a raw or defended personal-agent stack, where does it sit on the utility–security frontier? The expected practical lessons are: (1) do not deploy cross-boundary personal agents with raw or generic privacy prompts alone; (2) measure utility and security together, because refusal-only systems are safe but useless; (3) evaluate multi-turn interactions, because one-shot tests understate risk; and (4) treat relationship context as a security parameter, not just a personalization feature.

Toward architectural defences. The defense-frontier layer distinguishes prompt hardening from architectural isolation. If D3 and D4 improve security only incrementally while D5 changes the frontier, the conclusion is not merely that prompts need to be better. It is that personal-agent platforms need permissioned context construction: do not ask the agent to avoid sharing data it can see; ensure the data is not mounted for that relationship unless it should be available. D6 then tests whether escalation can recover utility lost through isolation.

Limitations. **Staged scope:** the benchmark is intentionally not a full factorial grid. This improves interpretability but means each layer answers a bounded question. **Synthetic scenario:** the personal agent’s data is synthetic, although it is structured to resemble realistic notes, memory, and tools. **Evaluation method:** gold-fact string matching may miss paraphrased leaks or trigger on coincidental matches; we will supplement it with LLM-based and human spot-check evaluation. **Model coverage:** Layer 4 tests five model families, but the benchmark should be rerun as new agent models and tool-use stacks emerge. **Attack coverage:** A0 and A2 cover natural and progressive multi-turn pressure; PAIR-style optimization and persuasion attacks remain benchmark extensions.

Table 9: Planned defense-frontier result table. Layers 3, 5, and 6 are not required for the minimum NeurIPS benchmark but define how PART-Bench scales into the thesis and follow-up defense paper.

Defence	Type	Utility	Security	Expected role
D0 No defence	Raw	XXXX	XXXX	Unsafe baseline
D1 Generic prompt	Prompt	XXXX	XXXX	Security theater baseline
D2 Category deny list	Prompt	XXXX	XXXX	First Pareto improvement
D3 Spotlighting	Prompt	XXXX	XXXX	Incremental prompt hardening
D4 Instruction hierarchy	Prompt	XXXX	XXXX	Stronger prompt-level ceiling
D5 MCC isolation	Architectural	XXXX	XXXX	Qualitative security jump
D6 MCC + escalation	Architectural	XXXX	XXXX	Utility recovery and fewer false refusals

Table 10: Planned action-safety track. Action tasks are separated from QA because the failure mode is unauthorized state change rather than information disclosure.

Task family	Authorized utility	Unauthorized action rate	Expected risk
Calendar	XXXX	XXXX	Exposes full calendar or schedules without permission
Notes	XXXX	XXXX	Shares, edits, or summarizes restricted notes
Messaging	XXXX	XXXX	Drafts or sends messages under the wrong principal
Todos	XXXX	XXXX	Creates or completes tasks for the wrong actor

Benchmark release. We will release the benchmark suite including: evaluation queries with gold key facts, agent configuration files (notes, memory, tools), the heartbeat evaluation engine, action-task templates, social-network setup templates, and evaluation scripts. The framework is model-agnostic and explicitly staged so future work can add models, attacks, defences, action families, and relationship graphs without changing the core utility–security contract.

References

- [1] Anthropic. Model context protocol: Connecting llms to external systems. <https://modelcontextprotocol.io>, 2024.
- [2] Patrick Chao, Alexander Robey, Edgar Dobriban, Hamed Hassani, George J Pappas, and Eric Wong. Jailbreaking black box large language models in twenty queries. In *IEEE Conference on Safe and Trustworthy Machine Learning (SaTML)*, 2025.
- [3] Cognition Labs. Devin: The first ai software engineer. <https://www.cognition.ai/devin>, 2024.
- [4] Edoardo DeBenedetti, Jie Zhang, Mislav Balunovic, Luca Beurer-Kellner, Marc Fischer, and Florian Tramer. Agentdojo: A dynamic environment to evaluate prompt injection attacks and defenses for llm agents. In *Advances in Neural Information Processing Systems*, volume 37, 2024.
- [5] Walid Gomaa, Ahmed Salem, and Sahar Abdelnabi. Converse: Benchmarking contextual safety in agent-to-agent conversations. *arXiv preprint arXiv:2511.05359*, 2025. EACL 2026 Findings.
- [6] Google. Agent-to-agent protocol: Enabling agent interoperability. <https://github.com/google/a2a-protocol>, 2025.
- [7] Taicheng Guo, Xiuying Chen, Yaqi Wang, Ruidi Chang, Shichao Pei, Nitesh V Chawla, Olaf Wiest, and Xiangliang Zhang. Large language model based multi-agents: A survey of progress and challenges. *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence*, 2024.

Table 11: Planned social-scaling track. These settings measure whether leakage is a property of one conversation or of the relationship network around the personal agent.

Setup	Metric	Expected pattern
S1 Single requester	Utility / security	Clean baseline for comparison
S3 Three requesters	Category leak by relationship	Different relationships pull different categories across the boundary
S4 Four interacting agents	Cross-agent leakage path	Agents may route around a boundary by asking each other
S50 Fifty requesters	Aggregate leak distribution	Social graph composition changes population-level leakage

- 451 [8] Keegan Hines, Gary Lopez, Matthew Hall, Federico Zarfati, Yonatan Zunger, and Emre Kici-
452 man. Defending against indirect prompt injection attacks with spotlighting. *arXiv preprint*
453 *arXiv:2403.14720*, 2024. Microsoft.
- 454 [9] Priyanshu Juneja, Lokesh Balaji Pasupulati, Alon Albalak, Wenyue Hua, and William Yang
455 Wang. Magpie: A benchmark for multi-agent contextual privacy evaluation. *arXiv preprint*
456 *arXiv:2510.15186*, 2025.
- 457 [10] Manus AI. Manus: Autonomous desktop agent. <https://www.manus.ai>, 2025.
- 458 [11] Meta AI. The rule of two: A security framework for ai agents. Meta AI Blog, 2025.
- 459 [12] Milad Nasr et al. The attacker moves second: Evaluating prompt injection defenses under
460 adaptive attack. *arXiv preprint*, 2025.
- 461 [13] OpenClaw Team. Openclaw: Open-source agent framework with skills marketplace. <https://github.com/openclaw>, 2025.
- 463 [14] Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G Patil, Ion Stoica,
464 and Joseph E Gonzalez. Memgpt: Towards llms as operating systems. *arXiv preprint*
465 *arXiv:2310.08560*, 2023.
- 466 [15] Jeonghwan Park, Seonghyeon Kim, Hyunju Ju, Junghyun Lim, Soyeon Choi, Ohyeon Kwon,
467 Joonho Kim, and Sangmin Yeo. Pac-bench: Evaluating multi-agent collaboration under privacy
468 constraints. *arXiv preprint arXiv:2604.11523*, 2026.
- 469 [16] Mark Russinovich, Ahmed Salem, and Ronen Eldan. Great, now write an article about that:
470 The crescendo multi-turn llm jailbreak attack. In *USENIX Security Symposium*, 2025.
- 471 [17] Timo Schick, Jane Dwivedi-Yu, Roberto Dessi, Roberta Raileanu, Maria Lomeli, Eric Hambro,
472 Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. Toolformer: Language models can
473 teach themselves to use tools. *Advances in Neural Information Processing Systems*, 36, 2023.
- 474 [18] Sander Schulhoff, Jeremy Pinto, Ansum Khan, Louis-François Bouchard, Chenglei Si,
475 Sidahmed Anati, Naty Tagber, Marzena Karpinska, Brieuc Dolan, and Jordan Boyd-Graber.
476 Hackaprompt: An adversarial prompt engineering competition. In *Findings of the Association*
477 *for Computational Linguistics: EMNLP 2023*, 2024.
- 478 [19] Natalie Shapira, Chris Wendler, Howard Yen, et al. Agents of chaos. *arXiv preprint*
479 *arXiv:2602.20021*, 2026.
- 480 [20] Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao.
481 Reflexion: Language agents with verbal reinforcement learning. *Advances in Neural Information*
482 *Processing Systems*, 36, 2023.
- 483 [21] Yashar Talebirad and Amirhossein Nadiri. Multi-agent collaboration: Harnessing the power of
484 intelligent llm agents. *arXiv preprint arXiv:2306.03314*, 2023.
- 485 [22] Sam Toyer, Olivia Watkins, Ethan Mendes, Justin Svegliato, Luke Bailey, Tiffany Wang, Isaac
486 Ong, Karim Elmaaroufi, Pieter Abbeel, Trevor Darrell, Alan Ritter, and Stuart Russell. Tensor
487 trust: Interpretable prompt injection attacks from an online game. In *Advances in Neural*
488 *Information Processing Systems*, volume 37, 2024.

- [23] Eric Wallace, Kai Xiao, Jan Leike, Lilian Weng, Johannes Heidecke, and Alex Beutel. The instruction hierarchy: Training llms to prioritize privileged instructions. *arXiv preprint arXiv:2404.13208*, 2024. OpenAI.
- [24] Lei Wang, Wanyu Xu, Yihuai Lan, Zhiqiang Hu, Yunshi Lan, Roy Ka-Wei Lee, and Ee-Peng Lim. Plan-and-solve prompting: Improving zero-shot chain-of-thought reasoning by large language models. *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, 2023.
- [25] Yuxuan Wang and Nan Jiang. Agentsocialbench: Evaluating privacy risks in human-centred agentic social networks. *arXiv preprint arXiv:2604.01487*, 2026.
- [26] Oussama Yagoubi, Gloria Badu-Marfo, and Raneem Al Mallah. Agentleak: A full-stack benchmark for privacy leakage in multi-agent llm systems. *arXiv preprint arXiv:2602.11510*, 2026.
- [27] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. React: Synergizing reasoning and acting in language models. *arXiv preprint arXiv:2210.03629*, 2023.
- [28] Yi Zeng, Hongpeng Lin, Jingwen Zhang, Diyi Yang, Ruoxi Jia, and Weiyan Shi. How johnny can persuade llms to jailbreak them: Rethinking persuasion to challenge ai safety by humanizing llms. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024.
- [29] Qiusi Zhan, Zhixiang Liang, Zifan Ying, and Daniel Kang. Injecagent: Benchmarking indirect prompt injections in tool-integrated llm agents. In *Findings of the Association for Computational Linguistics: ACL 2024*, 2024.
- [30] Andy Zou, Zifan Wang, J Zico Kolter, and Matt Fredrikson. Universal and transferable adversarial attacks on aligned language models. *arXiv preprint arXiv:2307.15043*, 2023.

A Observed Attack Patterns

Under D2 (category-specific deny list), residual leaks in single-step evaluation cluster into identifiable patterns:

Table 12: Distribution of D2 leak causes in single-step evaluation (XXXX total leaks).

Pattern	Count	Share
Category ambiguity (data accessible but semantically sensitive)	XXXX	XXXX%
Social framing (personal question in work-acceptable frame)	XXXX	XXXX%
Partial compliance failure (refusal then continued disclosure)	XXXX	XXXX%
Other	XXXX	XXXX%

Category ambiguity is the dominant failure mode: data stored in “Work” folders that is semantically sensitive (e.g., board meeting notes containing investor terms, HR documents with salary benchmarks). The agent’s policy correctly protects “personal finance” but does not recognise that a board memo discussing fundraising terms is financially sensitive.

B Experimental Configuration Details

Alex’s agent. System prompt establishes identity (Co-founder & CTO, Oxford CS background). Hundreds of notes and todos across 6 folder categories. Tools include search, retrieval, creation, completion, and scheduling interfaces (see Table 2). Memory contains preferences and interaction history.

525 **Tina's agent.** Operates from a HEARTBEAT.md instruction file. Phase 1 (information gathering):
526 systematically works through pending questions. Phase 2 (deeper probing): re-asks refused questions
527 with different framing. Maintains a MEMORY.md checklist tracking answered/refused/pending status
528 per question.

529 **Heartbeat configuration.** Each tick: Tina's agent selects 1–3 questions, sends a message, Alex's
530 agent responds. XXXX ticks per group in multi-turn mode.

NeurIPS Paper Checklist

The checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. Do not remove the checklist: **The papers not including the checklist will be desk rejected.** The checklist should follow the references and follow the (optional) supplemental material. The checklist does NOT count towards the page limit.

Please read the checklist guidelines carefully for information on how to answer these questions. For each question in the checklist:

- You should answer [Yes], [No], or [N/A].
- [N/A] means either that the question is Not Applicable for that particular paper or the relevant information is Not Available.
- Please provide a short (1–2 sentence) justification right after your answer (even for [N/A]).

The checklist answers are an integral part of your paper submission. They are visible to the reviewers, area chairs, senior area chairs, and ethics reviewers. You will also be asked to include it (after eventual revisions) with the final version of your paper, and its final version will be published with the paper.

The reviewers of your paper will be asked to use the checklist as one of the factors in their evaluation. While [Yes] is generally preferable to [No], it is perfectly acceptable to answer [No] provided a proper justification is given (e.g., error bars are not reported because it would be too computationally expensive” or “we were unable to find the license for the dataset we used”). In general, answering [No] or [N/A] is not grounds for rejection. While the questions are phrased in a binary way, we acknowledge that the true answer is often more nuanced, so please just use your best judgment and write a justification to elaborate. All supporting evidence can appear either in the main paper or the supplemental material, provided in appendix. If you answer [Yes] to a question, in the justification please point to the section(s) where related material for the question can be found.

IMPORTANT, please:

- **Delete this instruction block, but keep the section heading “NeurIPS Paper Checklist”.**
- **Keep the checklist subsection headings, questions/answers and guidelines below.**
- **Do not modify the questions and only use the provided macros for your answers.**

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the paper does not include experiments.

- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).

- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer **[N/A]** means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer **[N/A]** means that the paper does not include experiments.
- The authors should answer **[Yes]** if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer **[N/A]** means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.

- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the paper poses no such risks.

- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. Declaration of LLM usage

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.